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TITLE PAGE

Problem Statement ID – 26153

Problem Statement Title – AI based Network Attack Forecasting from Network Traffic Data

Theme – Blockchain & Cybersecurity

PS Category – Software

Organisation – National Technical Research Organisation (NTRO)

Team ID – [Team ID]

Team Name – Git with It

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PREDICTIVE CYBER DEFENCE

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We do not classify traffic — we learn how a network's state evolves, then run that model forward to see where a host is heading.

Proposed Solution

- Traffic → 60-second windows; each becomes a 39-feature snapshot of one host
- A world model learns the transition: what state usually follows this state
- Stop feeding it traffic — it generates the next 10 states on its own
- Each imagined state is scored: infiltration risk + MITRE ATT&CK stage

How It Addresses the Problem

- Answers “where is this host heading”, not “was that flow bad”
- 10-minute lookahead — acts before the kill chain closes
- Flow-level and packet-level features together: TTL variance, TCP window, retransmissions, scan signatures
- Every alert carries three explanations — no black box

Innovation and Uniqueness

- World model, not a classifier — the heads read the latent state, so they still work on imagined states
- Provable: an imagination-only loss yields zero encoder gradient (asserted in our test suite)
- ATT&CK stages derived from CTU-13's own flow annotations, not hand-waved
- Runs fully offline on CPU — 138k parameters

Working prototype today • 95,518 network states from 2.8M real flows • F1 0.984 vs 0.744 for the strongest baseline • 149 hosts triaged in 0.5 s • minute-by-minute replay of the forecast

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TECHNICAL APPROACH

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Ingest
NetFlow / PCAP (Scapy)

Feature extraction
39 flow + 23 packet features

Network state
60-second (host, window) cells

World model
Encoder → GRU → prior / posterior

Forward simulation
10 imagined states + uncertainty band

Decision support
Risk • ATT&CK stage • explanations

Technologies Used

- Python 3.13, PyTorch (CPU-only)
- Scapy — raw PCAP parsing, flow reconstruction
- pandas / NumPy — vectorised feature pipeline
- scikit-learn — baselines and metrics
- FastAPI + zero-dependency HTML dashboard
- 100% open source • no cloud, no external API

Model — Recurrent State-Space

- Encoder compresses each window into a latent state
- GRU carries context; a prior predicts the next latent without seeing traffic
- Decoder reconstructs traffic, so the latent must encode real network state
- Trained on its own rollouts, so multi-step forecasting is learned, not hoped for
- Causal attention — masked so no step sees its future

Methodology and Validation

- Dataset: CTU-13 (13 real botnet captures, CC BY)
- Split by time, never randomly — train on each capture's past, test on its future, with a guard band
- Baselines given identical features, scaler and labels
- Deterministic inference + seeded rollouts — the benchmark reproduces byte for byte
- Test suites for model shapes and PCAP header parsing

Offline dashboard: minute-by-minute replay of the forecast • kill-chain timeline • network topology • three explanation channels per prediction

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FEASIBILITY AND VIABILITY

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Feasibility — already built and measured

Sustained compromise, temporal split	F1	FPR	Recall
World model (ours)	**0.984**	**0.000**	**1.000**
Logistic regression, 8-window	0.744	0.004	0.912
Logistic regression, 1 window	0.550	0.009	0.842

Why it is deployable

- Runs on a laptop CPU — no GPU, no cloud, no licence cost
- Consumes NetFlow/IPFIX that enterprises already export
- Sub-second triage of a full capture after first load
- Fully offline — suitable for air-gapped Critical Information Infrastructure
- MITRE stage macro-F1 0.455 vs 0.353; beats a persistence baseline at 9 of 10 forecast horizons
- Replay reconstructs what the model knew at every minute, so an analyst can audit a decision after the fact

Potential Challenges and Risks → Our Mitigation

Label leak through metadata

The available PCAP held only botnet traffic, so a "packet data present" flag became a perfect label proxy → detected it, removed it; F1 rose 0.23 → 0.98

Dataset has no pre-infection baseline

CTU-13 hosts are malicious in every window → we forecast kill-chain progression instead, and say so plainly

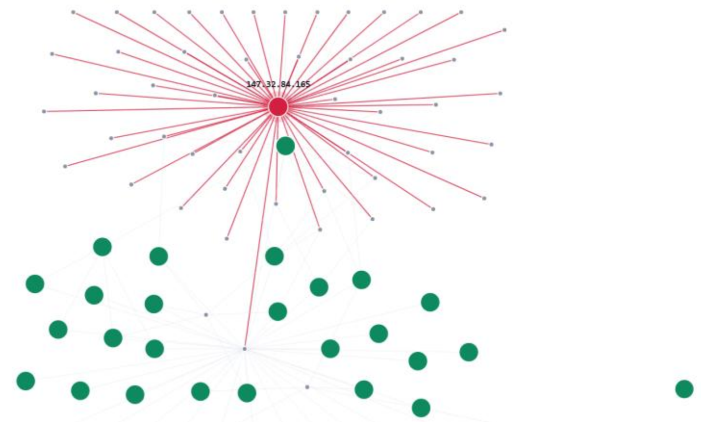
Generalising to unseen malware families

Only ~4 infected hosts to learn from → reported as a measured limitation, not hidden; needs more captures

Short-burst compromise is missed

Same IP scores 1.000 with 284 malicious windows and 0.000 with 20 → not memorisation, a dependence on sustained activity; F1 above is for sustained compromise

Live output — the compromised host and its malicious fan-out



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IMPACT AND BENEFITS

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Shifting the defender from reacting after compromise to acting during the kill chain

Who Benefits

- SOC analysts — a ranked queue instead of an alert flood
- Critical Information Infrastructure: power, banking, telecom, transport
- CERT-In and national response teams
- Enterprises already exporting NetFlow, with no new hardware to buy

Operational Benefits

- Up to 10 minutes of warning before the kill chain closes
- False positive rate 0.000 on the held-out period — analyst time is not wasted
- Replay lets an analyst rewind an incident minute by minute and see what the model knew, when
- Explanations make alerts auditable and challengeable
- ATT&CK stage names map onto existing playbooks

Strategic and Economic

- Fully indigenous, open-source — no foreign vendor dependency for national infrastructure
- Zero licence cost; runs on existing hardware
- Air-gap friendly — nothing leaves the network
- Earlier detection cuts breach cost, downtime and data loss

Every claim on this deck is reproducible from the repository: one command prepares the data, one trains, one benchmarks.

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RESEARCH AND REFERENCES

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Datasets and Knowledge Bases

CTU-13 Dataset — Stratosphere IPS, CTU Prague

García et al., “An empirical comparison of botnet detection methods”, Computers & Security, 2014

mcfp.felk.cvut.cz/publicDatasets/CTU-13-Dataset/

MITRE ATT&CK Enterprise Matrix

Tactics TA0043, TA0001, TA0008, TA0011, TA0010

attack.mitre.org

CIC-IDS2017 / CSE-CIC-IDS2018 — UNB

Reviewed; access now gated behind registration

NCIIPC — nciipc.gov.in

Methods and Literature

World models / latent dynamics

Ha & Schmidhuber, “World Models”, NeurIPS 2018

Hafner et al., “Learning Latent Dynamics for Planning from Pixels” (PlaNet/RSSM), ICML 2019

Hafner et al., “Dream to Control” (Dreamer), ICLR 2020

Explainability

Sundararajan et al., “Axiomatic Attribution for Deep Networks” (Integrated Gradients), ICML 2017

Vaswani et al., “Attention Is All You Need”, NeurIPS 2017

Tools

Scapy • PyTorch • scikit-learn • FastAPI